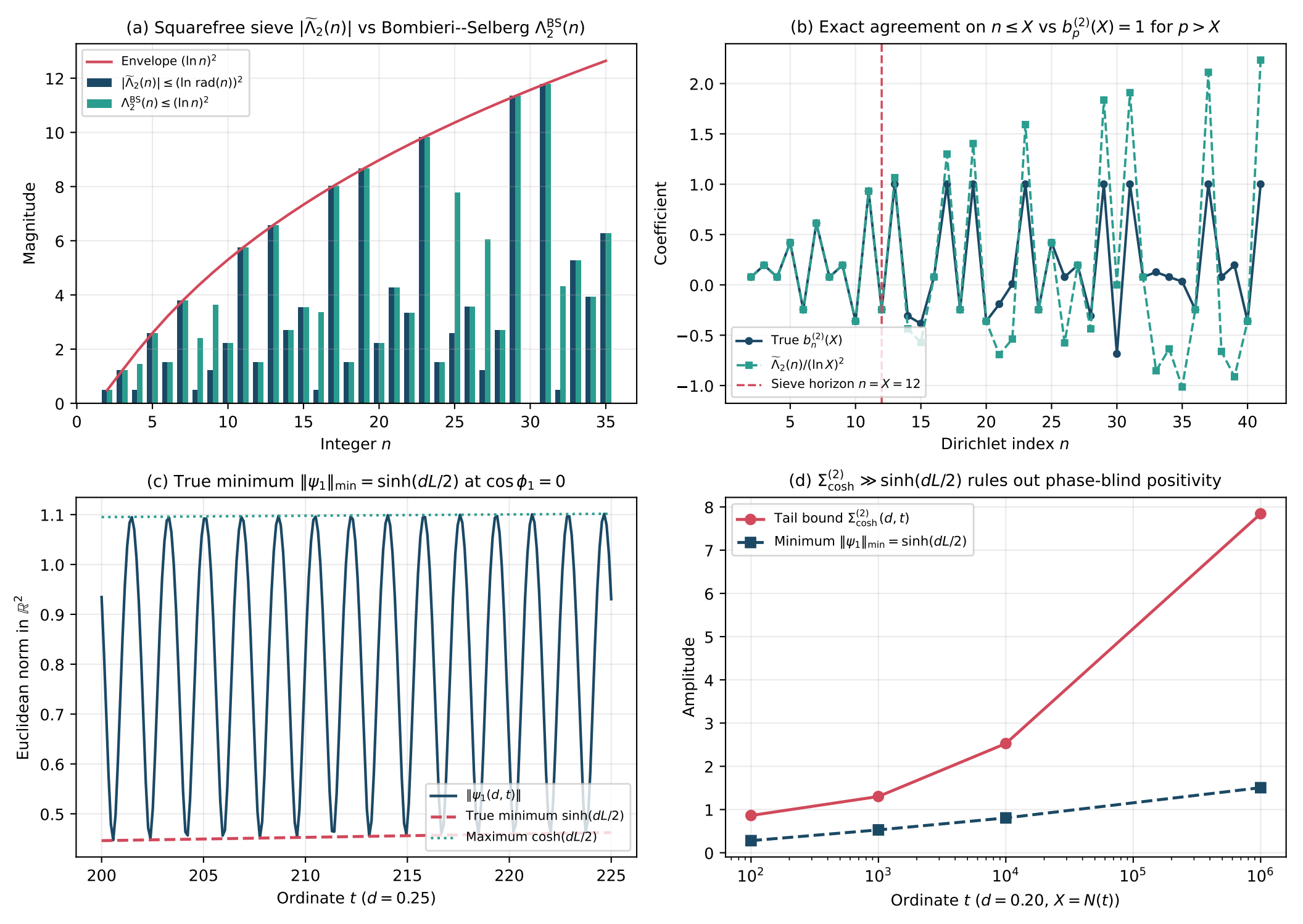
(a) Squarefree sieve $|\widetilde{\Lambda}_2(n)|$ vs Bombieri--Selberg $\Lambda_2^{\mathrm{BS}}(n)$
Envelope $(\ln n)^2$
$|\widetilde{\Lambda}_2(n)| \le (\ln \mathrm{rad}(n))^2$
$\Lambda_2^{\mathrm{BS}}(n) \le (\ln n)^2$
Magnitude
Integer n
(b) Exact agreement on $n \le X$ vs $b_p^{(2)}(X) = 1$ for $p > X$
True $b_n^{(2)}(X)$
$\widetilde{\Lambda}_2(n)/(\ln X)^2$
Sieve horizon $n = X = 12$
Coefficient
Dirichlet index n
(c) True minimum $\|\psi_1\|_{\min} = \sinh(dL/2)$ at $\cos\phi_1 = 0$
$\|\psi_1(d,t)\|$
True minimum $\sinh(dL/2)$
Maximum $\cosh(dL/2)$
Euclidean norm in $\mathbb{R}^2$
Ordinate t $(d = 0.25)$
(d) $\Sigma_{\cosh}^{(2)} \gg \sinh(dL/2)$ rules out phase-blind positivity
Tail bound $\Sigma_{\cosh}^{(2)}(d,t)$
Minimum $\|\psi_1\|_{\min} = \sinh(dL/2)$
Amplitude
Ordinate t $(d = 0.20,\ X = N(t))$